



Math Level 4 explores topics such as multiplication and division of multi-digit numbers, addition and multiplication of fractions, special two-dimensional shapes, symmetry, logic, factors and prime numbers, and probability. Problem-solving strategies are emphasized throughout the course, and each lesson includes challenging word problems and puzzles that guide students from the basics toward a deep understanding of the material.

Beast Academy provides a fun, rigorous, and challenging complete curriculum for aspiring math beasts. Our philosophy is that students develop more by learning to solve problems they haven't seen before, as opposed to offering repeated drills that students can memorize their way through.

Book: Beast Academy 4A – 4D

Time Commitment: This 36-week course includes 105 minutes in-class and 60-90 minutes of homework each week, corresponding to a full year course.

Content:

Week	Topic
1	Shapes and Angles
2	Parallel and Perpendicular
3	The Distributive Property
4	The Multiplication Algorithm
5	The Units Digit
6	Introduction to Exponents
7	Perfect Squares as Powers
8	Binary Numbers
9	Test on Book 4A
10	Counting Lists & Choices
11	Venn Diagrams & Factorials
12	Counting Pairs
13	Special Quotients
14	Long Division
15	Divisibility Rules
16	Logic Part 1
17	Logic Part 2
18	Test on Book 4B

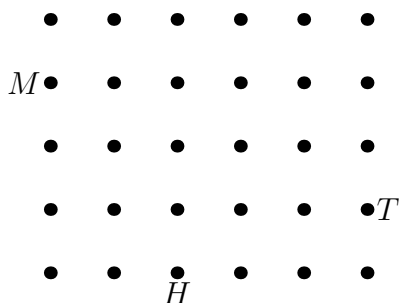
Week	Topic
19	Factors and Prime Numbers
20	The Sieve and Divisibility
21	Prime Factorization
22	Adding Fractions
23	Mixed Numbers Part 1
24	Subtracting Fractions
25	Negative Numbers
26	Adding and Subtracting Integers
27	Test on Book 4C
28	Multiplication by a Fraction
29	Mixed Numbers Part 2
30	Dividing by a Fraction
31	Introduction to Decimals
32	Comparing Decimals
33	Adding & Subtracting Decimals
34	Computing Probability
35	Probability Grab Bag
36	Test on Book 4D

Common Core State Standards:

The content covered in our Math Level 4 class is loosely based on the standard created by the Common Core State Standards Initiative. We cover many of the standards from grade 4, as well as standards for grades 5, 6, 7, and 8. You can find a document with the complete list on our Beast Academy Planning page (<https://beastacademy.com/resources/planning>). Or, you can access the standards by navigating directly to <https://beastacademy.com/u/pdfs/planning/standards/BA4Standards.pdf>

Sample Problems:

- Label point A on the diagram to create parallelogram $MATH$.



- What is the units digit of 43^{100} ?
- The base-10 number 39 is written in base-2 as 100,111. How is the base-10 number 78 written in base-2?
- What is the smallest number that has five different prime factors?
- Which integer is the same distance from 18 as it is from -4 on the number line?
- What is the value of the expression below?

$$2 + 4 + 6 + \cdots + 996 + 998 + 1,000 - 1 - 3 - 5 - \cdots - 995 - 997 - 999?$$

- Place a decimal point between two digits of the number below so that the resulting number has tens digit 9 and tenths digit 8.

9 8 9 9 8

- Two of the digits of 7,861 are selected at random and multiplied. What is the probability that their product is odd?

